

Model fixes and the GP / embedding comparison

Five fixes found while comparing the paper's embedding systems, and what the comparison shows

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1 Summary

- Five defects were found and fixed while setting up the comparison. Fixes 1, 2, 4 and 5 were also present in the code the paper's hardware flights used (tag `archive/main-2026-05-02`), and in the code of the paper's SITL runs.
- **The GP decides whether the certificate is true; the embedding system barely matters here.** With the time-varying GP every embedding system keeps the vehicle inside its certified tube (0 % of the time outside in the numerical runs); with a time-invariant GP every embedding system is violated about 40 % of the time, by up to 0.29 m.
- With the time-varying GP the three embedding systems certify the same horizon with the same tubes; (66)-(67) is the cheapest (about a third less compute per rollout).

2 The five fixes

	what was wrong	effect	fix
1	the embedding used $+[M_u]K$, but the controller applies $u = u_r - K(x - x_r)$	the tube modelled the wrong closed loop (looser than needed)	$K_p = -K$ in (66)-(69); certifies 0.45-0.62 s instead of 0.40-0.53 s on the same plans
2	the dynamics' Jacobian was taken over $w \in \mu(x_r) \pm s$ only	the $\partial/\partial\theta$ entries depend on w ; the domain must contain every disturbance the box can produce	domain $[\underline{\gamma}, \overline{\gamma}]$ over the box, for all variants

	what was wrong	effect	fix
3	<code>RolloutPlan.index_at</code> truncated exact row times ($0.03/0.01 = 2.999 \dots \rightarrow 2$)	45 % of numerical ticks used the previous row (controller and metrics)	$+10^{-6}$ row tolerance (Python and C++)
4	the planar state's h, v were filled with inertial (v_y, v_z); the model defines them in the body frame	up to 1.7 m/s off at 30° roll, in the tube's start state and the RTA feedback	$[h; v] = R(\theta)^\top [v_y; v_z]$ (Python and C++)
5	the state came from the <code>mocap_px4_relays</code> relay: a 25 ms timer behind a 50 Hz rate gate that local position (exactly 50 Hz) keeps failing	25 Hz with gaps up to 150 ms, i.e. stale state	the node and the C++ loop read <code>/fmu/out/vehicle_odometry</code> (100 Hz) directly

Effect of fixes 4 and 5 on a default SITL flight:

	before	+ body-frame h, v	+ 100 Hz PX4 odometry
tracking RMSE lateral / vertical	88 / 67 mm	28 / 43 mm	20 / 32 mm
worst position outside the certified tube	0.45 m	0.21 m	0.11 m
C++ fast loop, RMSE lateral / vertical	86 / 50 mm		23 / 28 mm

New metric: escapes. `analysis.tube_escapes` measures, at every certified control tick, how far the vehicle's position is outside the certified box of the row in use: the true state in the numerical simulation, the EKF2 estimate in SITL and on hardware. A certificate that is correct has no escapes beyond the 10 ms integration step of the tube (about 1 mm).

Still open. In SITL, the TV-GPR variants still leave the tube about 20 % of the time (up to 0.12 m), concentrated in the fast descent after the RTA takes over. The excursion correlates with speed; aerodynamic drag is weak in the estimated wind (-0.08 N per m/s), and the steady -1 N in the vertical wind is the thrust-map offset of doc 04. The remaining mismatch is between the planar model and the full PX4 vehicle (rate-loop dynamics, out-of-plane motion); it affects every variant equally.

3 The comparison

Two wind models (TV-GPR, $\epsilon = 0.25$; GPR = TVGPR with $\epsilon = 0$) $\times$ the paper's three embedding systems (Appendix A), all at the same certificate: position within 0.25 m of the reference, 0.3 m floor, $\delta = 0$. Numerical closed loop: 10 cases per variant (calm, paper winds $\times 0.3$ / $\times 0.6$ / $\times 1.0$ with 3 seeds), backup off. PX4 SITL: 3 flights per variant, backup on.

Time outside the certified tube (numerical wind runs, true state · SITL, EKF2 estimate):

embedding system	TV-GPR	GPR
(68)-(69): first order in u and w	A: 0 % · 19 % (default)	D: 41 % · 65 %
(66)-(67): first order in u	E: 0 % · 20 %	C: 40 % · 67 %
(64)-(65): no first-order terms	F: 0.03 % · 20 %	B: 40 % · 71 %

	A	E	F	D	C	B
numerical: worst excursion	0.9 mm	0.9 mm	1.2 mm	0.10 m	0.29 m	0.28 m
numerical: certified horizon (median)	0.59 s	0.59 s	0.59 s	0.78 s	0.80 s	0.70 s
numerical: tube half-width at +0.25 s	26 mm	26 mm	25 mm	2.6 mm	2.5 mm	2.4 mm
numerical: rollout	6.9 ms	4.3 ms	6.8 ms	7.0 ms	6.1 ms	6.2 ms
SITL: worst excursion	0.12 m	0.12 m	0.15 m	0.27 m	0.80 m	0.26 m
SITL: tracking RMSE lateral / vertical (mm)	20 / 27	20 / 26	20 / 30	24 / 32	27 / 42	27 / 42

Why the GP matters. The time-invariant GP’s uncertainty stays small near its 9 observations, so its tubes are about $10\times$ thinner and certify further ahead; but the wind drifts over time, so the GP is confidently wrong and the real vehicle leaves the “certified” tube. The TV-GPR’s forgetting grows the uncertainty over the look-ahead, which is exactly what keeps the tube honest.

Why the embedding barely matters. With TV-GPR the growth of σ along the look-ahead dominates the tube’s growth, so the first-order disturbance term of (68)-(69) buys no measurable tightening over (66)-(67); and for this vehicle (64)-(65) coincides with (66)-(67), because the inputs enter through a constant selector matrix: $[M_u](K[e]) = ([M_u]K)[e]$. A benefit of the first-order terms would need a disturbance whose mean varies across the tube much more than its uncertainty grows along it.

Data, tables (GP $\times$ embedding grids, two-way effect split), per-flight results and figures: `packages/px4_rta_mm_gpr/scripts/data_analysis/05_embedding_comparison.ipynb`. Reproduce with `--gp tv|static --embedding uw|u|none (node) or SimConfig(gp_epsilon=..., embedding=...)`.